

Robotic Ankle Joint Actuator Sizing & Comprehensive Parameter Sweep Report

Complete autonomous analysis of standard vs orthogonal mounting layouts sweeping ankle spread, motor depth offset, and attachment offset for Robstride RS02, RS03, and RS04 actuators.

Prepared For: Robotic Design Review & Ankle Actuator Selection

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Evaluation Focus: Minimum Guaranteed Torques and Discrete Joint Angles

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1. Executive Summary & Sizing Methodology

To size the ankle joint actuators for roll and pitch stability, we sweep the joint workspace under two main scenarios: the current robot design (narrow base) and a parallel mechanism (widened base). The sizing maps motor rated and peak limits to ankle output torque capacity using the analytical Jacobian.

For each design, we evaluate standard parallel mounting against decoupled orthogonal mounting. To guarantee safety and performance, the analysis sweeps variations in motor placement depth and linkage attachment offset.

1.1 Kinematics Calculation Method & Formulas

The linkage is modeled as a closed-loop parallel kinematic system. The position of each servo motor's crank tip c_i is constrained by the length of the pushrod L_i to the footplate attachment joint a_i :

Linkage Length Constraint:
$$||c_i(\theta_i) - a_i(p, r)||^2 = L_i^2 \text{ (for } i = 1, 2)$$

Where:

- θ_i : Angular position of Motor i .
- p, r : Pitch and Roll orientation of the footplate.
- $c_i(\theta_i)$: Position of the crank tip in the global chassis frame.
- $a_i(p, r)$: Position of the pushrod attachment joint on the footplate in the global chassis frame:

$$a_i = a_c + R_x(p) \cdot R_y(r) \cdot a_{ib}$$

where a_c is the central Universal Joint coordinate and a_{ib} is the attachment joint position in the foot's local frame.

1.2 Jacobian and Torque Mapping

Differentiating the constraint equations with respect to time yields the system Jacobian J , which maps footplate velocities $[\dot{p}, \dot{r}]^T$ to motor velocities $[\dot{\theta}_1, \dot{\theta}_2]^T$:

Velocity Mapping: $\dot{\theta} = J \cdot \dot{x}$
Jacobian Element (Row i , Col j):
$$J_{ij} = ((c_i - a_i) \cdot (\partial a_i / \partial x_j)) / ((c_i - a_i) \cdot (\partial c_i / \partial \theta_i))$$

By the Principle of Virtual Work, the torque transmission is mapped as:

Torque Mapping: $\tau_{ankle} = J^T \cdot \tau_{motor}$

For a given motor torque limit (Rated/Peak), the ankle output torque is determined as the sum of contributions under motor limits, finding the worst-case minimum capacity across the Pitch (from -25° to +25°) and Roll (from -40° to +40°) sweeps.

2. Robstride RS02: Rated Sizing Sweep (Limit: 6.0 N·m)

RS02 Sizing Dimensions: Crank 1/2 Radius = 2.5 cm, Pushrod 1 Length = 24.3 cm, Universal Joint to Motor 2 Height = 16.0 cm, Foot to Universal Joint Height = 3.6 cm, Universal Joint to Motor 1 Height = 23.5 cm, Pushrod 2 Length = 16.3 cm.

2.1 Standard Parallel Layout (0° Yaw) Sweep

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Min Pitch Torque (N·m)	Critical Pitch Angle	Min Roll Torque (N·m)	Critical Roll Angle
2.3	3.2	-1.0	15.45	-6.0°	4.61	+40.0°
2.3	3.2	0.0	14.67	-25.0°	4.44	+40.0°
2.3	3.2	1.0	16.03	-25.0°	4.25	+40.0°
2.3	3.8	-1.0	18.44	-1.0°	4.61	+40.0°
2.3	3.8	0.0	17.82	-16.5°	4.44	+40.0°
2.3	3.8	1.0	18.58	-25.0°	4.25	+40.0°
2.3	4.4	-1.0	21.35	+1.0°	4.61	+40.0°
2.3	4.4	0.0	20.83	-12.5°	4.44	+40.0°
2.3	4.4	1.0	21.35	-24.5°	4.25	+40.0°
5.0	3.2	-1.0	15.71	-2.0°	10.92	+40.0°
5.0	3.2	0.0	14.95	-21.5°	11.90	+33.0°
5.0	3.2	1.0	16.03	-25.0°	14.90	+10.5°
5.0	3.8	-1.0	18.65	+2.0°	10.92	+40.0°
5.0	3.8	0.0	18.03	-13.5°	11.90	+33.0°
5.0	3.8	1.0	18.68	-25.0°	14.90	+10.5°
5.0	4.4	-1.0	21.53	+3.0°	10.92	+40.0°
5.0	4.4	0.0	21.00	-10.0°	11.90	+33.0°
5.0	4.4	1.0	21.53	-22.5°	14.90	+10.5°

2.2 Orthogonal Decoupled Layout (90° Yaw) Sweep

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Min Pitch Torque (N·m)	Critical Pitch Angle	Min Roll Torque (N·m)	Critical Roll Angle
2.3	3.2	-1.0	10.56	-10.5°	9.16	-34.0°
2.3	3.2	0.0	10.16	-25.0°	9.16	-40.0°
2.3	3.2	1.0	11.99	-25.0°	11.23	-40.0°
2.3	3.8	-1.0	12.20	-2.5°	11.59	-15.5°
2.3	3.8	0.0	11.80	-14.5°	11.26	-28.0°
2.3	3.8	1.0	13.34	-25.0°	12.71	-37.0°
2.3	4.4	-1.0	13.70	+0.5°	13.31	-8.0°
2.3	4.4	0.0	13.31	-10.0°	12.93	-18.5°
2.3	4.4	1.0	14.78	-21.5°	14.19	-28.5°
5.0	3.2	-1.0	14.00	-4.5°	13.56	-20.0°
5.0	3.2	0.0	13.28	-24.5°	12.48	-40.0°
5.0	3.2	1.0	14.64	-25.0°	13.87	-40.0°
5.0	3.8	-1.0	15.53	+1.5°	15.28	-7.0°
5.0	3.8	0.0	15.01	-14.5°	14.52	-25.0°
5.0	3.8	1.0	16.05	-25.0°	15.43	-36.5°
5.0	4.4	-1.0	16.98	+3.5°	16.80	-3.0°
5.0	4.4	0.0	16.57	-10.0°	16.15	-17.0°
5.0	4.4	1.0	17.62	-23.0°	17.00	-29.0°

2.3 Robstride RS02: Peak Sizing Sweep (Limit: 17.0 N·m)

Standard Layout (0° Yaw) Peak Sweep

Spread (E) (cm)	Y-Offset (L) (cm)	Attach (h) (cm)	Min Pitch Torque (N·m)	Min Roll Torque (N·m)
2.3	3.8	-1.0	52.26	13.07
2.3	3.8	0.0	50.50	12.59
2.3	3.8	1.0	52.65	12.04
5.0	3.8	-1.0	52.83	30.95
5.0	3.8	0.0	51.08	33.72
5.0	3.8	1.0	52.93	42.23

Orthogonal Layout (90° Yaw) Peak Sweep

Spread (E) (cm)	Y-Offset (L) (cm)	Attach (h) (cm)	Min Pitch Torque (N·m)	Min Roll Torque (N·m)
2.3	3.8	-1.0	34.58	32.83
2.3	3.8	0.0	33.44	31.90
2.3	3.8	1.0	37.80	36.01
5.0	3.8	-1.0	44.00	43.30
5.0	3.8	0.0	42.52	41.15
5.0	3.8	1.0	45.48	43.72

2.4 RS02 Discrete Joint Angle Torque Matrices (L = 3.8 cm, h = 0.0 cm)

Rated Limit (6.0 N·m) Torque Output

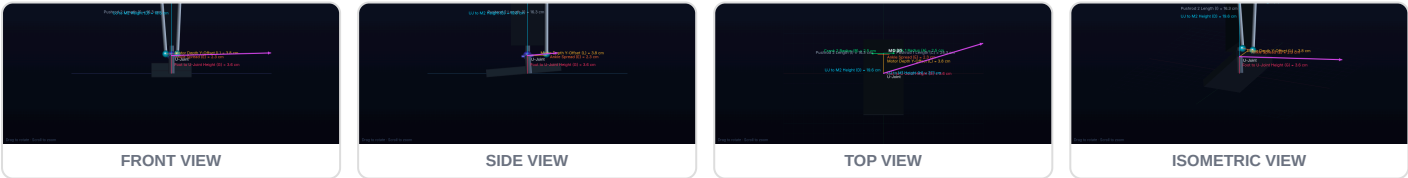
Pitch Angle (Roll=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-25.0°	18.21	18.77	12.13	15.44
-15.0°	17.83	18.04	11.80	15.01
-5.0°	18.38	18.33	11.97	15.23
0.0°	18.98	18.79	12.19	15.51
+5.0°	19.88	19.50	12.49	15.90
+15.0°	23.35	22.20	13.51	17.15
+25.0°	40.47	31.87	15.90	19.67

Peak Limit (17.0 N·m) Torque Output

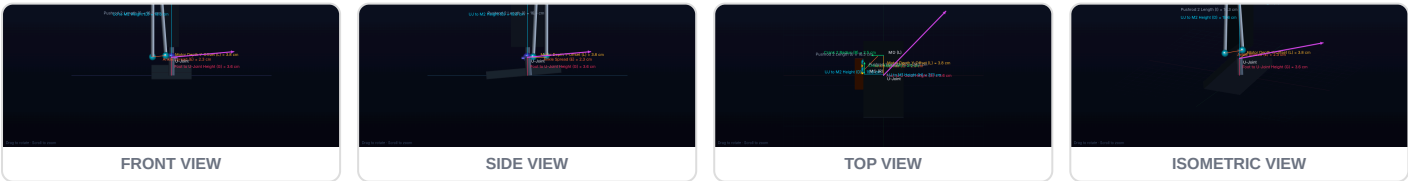
Roll Angle (Pitch=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-40.0°	13.99	53.86	33.19	43.37
-20.0°	15.93	37.46	32.21	41.28
-10.0°	16.29	35.92	33.37	42.27
0.0°	16.28	35.03	35.53	44.34
+10.0°	15.91	34.43	39.81	48.36
+20.0°	15.19	34.00	52.62	58.57
+40.0°	12.59	33.87	—	—

2.5 Annotated RS02 Kinematics Renders (with Dimension Callouts)

Configuration 1: Standard Layout (Ankle Spread E = 2.3 cm)



Configuration 2: Orthogonal Layout (Ankle Spread E = 2.3 cm)



3. Robstride RS03: Rated Sizing Sweep (Limit: 20.0 N·m)

RS03 Sizing Dimensions: Crank 1/2 Radius = 2.5 cm, Pushrod 1 Length = 22.2 cm, Universal Joint to Motor 2 Height = 11.2 cm, Foot to Universal Joint Height = 3.6 cm, Universal Joint to Motor 1 Height = 22.2 cm, Pushrod 2 Length = 11.2 cm.

3.1 Standard Parallel Layout (0° Yaw) Sweep

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Min Pitch Torque (N·m)	Critical Pitch Angle	Min Roll Torque (N·m)	Critical Roll Angle
2.3	3.6	-1.0	59.39	+11.0°	15.61	+40.0°
2.3	3.6	0.0	57.25	-4.5°	14.36	+40.0°
2.3	3.6	1.0	59.39	-20.0°	12.19	+39.5°
2.3	4.2	-1.0	68.69	+10.0°	15.61	+40.0°
2.3	4.2	0.0	66.84	-3.5°	14.36	+40.0°
2.3	4.2	1.0	68.69	-16.5°	12.19	+39.5°
2.3	4.8	-1.0	78.04	+9.0°	15.61	+40.0°
2.3	4.8	0.0	76.42	-2.5°	14.36	+40.0°
2.3	4.8	1.0	78.04	-14.5°	12.19	+39.5°
5.0	3.6	-1.0	59.78	+15.0°	43.68	+0.0°
5.0	3.6	0.0	57.60	+0.0°	40.00	-35.5°
5.0	3.6	1.0	59.78	-16.0°	43.61	+0.0°
5.0	4.2	-1.0	69.08	+13.0°	43.68	+0.0°
5.0	4.2	0.0	67.20	+0.0°	40.00	-35.5°
5.0	4.2	1.0	69.08	-13.5°	43.61	+0.0°
5.0	4.8	-1.0	78.45	+11.5°	43.68	+0.0°
5.0	4.8	0.0	76.80	+0.0°	40.00	-35.5°
5.0	4.8	1.0	78.45	-12.0°	43.61	+0.0°

3.2 Orthogonal Decoupled Layout (90° Yaw) Sweep

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Min Pitch Torque (N·m)	Critical Pitch Angle	Min Roll Torque (N·m)	Critical Roll Angle
2.3	3.6	-1.0	40.51	+4.0°	41.11	+6.5°
2.3	3.6	0.0	37.70	-6.5°	37.92	-2.5°
2.3	3.6	1.0	41.04	-16.0°	40.85	-10.5°
2.3	4.2	-1.0	45.20	+6.0°	45.76	+8.0°
2.3	4.2	0.0	42.50	-4.5°	42.71	-2.0°
2.3	4.2	1.0	45.51	-14.5°	45.39	-11.5°
2.3	4.8	-1.0	49.86	+6.5°	50.40	+8.0°
2.3	4.8	0.0	47.28	-3.5°	47.50	-1.5°
2.3	4.8	1.0	50.03	-13.0°	49.95	-11.0°
5.0	3.6	-1.0	51.45	+15.5°	51.46	+16.0°
5.0	3.6	0.0	48.80	+0.0°	48.80	+0.0°
5.0	3.6	1.0	51.44	-16.5°	51.44	-16.0°
5.0	4.2	-1.0	56.12	+13.5°	56.12	+13.5°
5.0	4.2	0.0	53.60	+0.0°	53.60	+0.0°
5.0	4.2	1.0	56.11	-14.0°	56.11	-14.0°
5.0	4.8	-1.0	60.82	+12.0°	60.83	+12.0°
5.0	4.8	0.0	58.40	+0.0°	58.40	+0.0°
5.0	4.8	1.0	60.81	-12.0°	60.81	-12.0°

3.3 Robstride RS03: Peak Sizing Sweep (Limit: 60.0 N·m)

Standard Layout (0° Yaw) Peak Sweep

Spread (E) (cm)	Y-Offset (L) (cm)	Attach (h) (cm)	Min Pitch Torque (N·m)	Min Roll Torque (N·m)
2.3	4.2	-1.0	206.06	46.84
2.3	4.2	0.0	200.53	43.07
2.3	4.2	1.0	206.06	36.56
5.0	4.2	-1.0	207.23	131.04
5.0	4.2	0.0	201.60	120.00
5.0	4.2	1.0	207.23	130.83

Orthogonal Layout (90° Yaw) Peak Sweep

Spread (E) (cm)	Y-Offset (L) (cm)	Attach (h) (cm)	Min Pitch Torque (N·m)	Min Roll Torque (N·m)
2.3	4.2	-1.0	135.61	137.27
2.3	4.2	0.0	127.50	128.14
2.3	4.2	1.0	136.53	136.17
5.0	4.2	-1.0	168.36	168.37
5.0	4.2	0.0	160.80	160.80
5.0	4.2	1.0	168.33	168.33

3.4 RS03 Discrete Joint Angle Torque Matrices (L = 4.2 cm, h = 0.0 cm)

Rated Limit (20.0 N·m) Torque Output

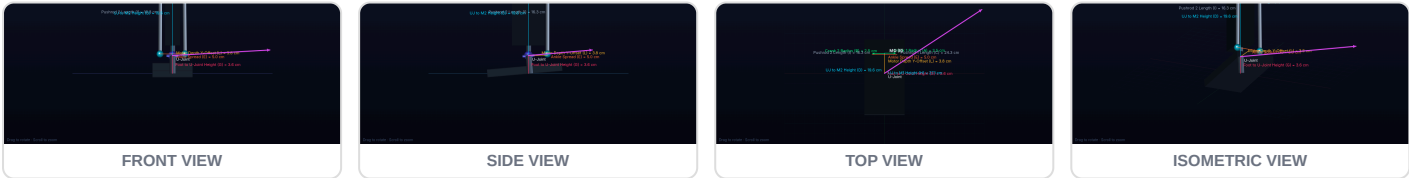
Pitch Angle (Roll=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-25.0°	84.95	93.96	50.90	68.35
-15.0°	70.32	72.90	43.91	56.59
-5.0°	66.90	67.70	42.50	53.85
0.0°	67.06	67.20	42.68	53.60
+5.0°	68.19	67.65	43.29	53.82
+15.0°	73.82	71.28	46.03	55.51
+25.0°	88.23	80.58	52.73	59.40

Peak Limit (60.0 N·m) Torque Output

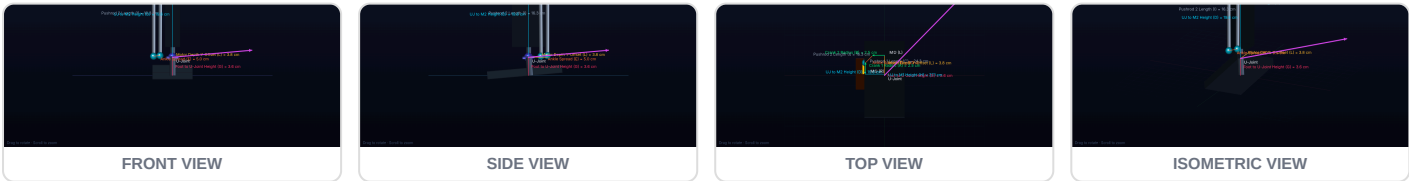
Roll Angle (Pitch=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-30.0°	49.58	120.00	188.24	248.33
-20.0°	52.78	120.00	140.71	178.19
-10.0°	54.61	120.00	130.04	164.00
0.0°	55.08	120.00	128.22	160.80
+10.0°	54.21	120.00	131.76	163.57
+20.0°	51.97	120.00	143.12	173.41
+30.0°	48.28	120.00	182.58	202.45

3.5 Annotated RS03 Kinematics Renders (with Dimension Callouts)

Configuration 3: Standard Layout (Ankle Spread E = 5.0 cm)



Configuration 4: Orthogonal Layout (Ankle Spread E = 5.0 cm)



4. Robstride RS04: Rated Sizing Sweep (Limit: 40.0 N·m)

RS04 Sizing Dimensions: Crank 1/2 Radius = 2.5 cm, Pushrod 1 Length = 22.2 cm, Universal Joint to Motor 2 Height = 11.2 cm, Foot to Universal Joint Height = 3.6 cm, Universal Joint to Motor 1 Height = 22.2 cm, Pushrod 2 Length = 11.2 cm.

4.1 Standard Parallel Layout (0° Yaw) Sweep

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Min Pitch Torque (N·m)	Critical Pitch Angle	Min Roll Torque (N·m)	Critical Roll Angle
2.3	3.6	-1.0	118.78	+11.0°	31.23	+40.0°
2.3	3.6	0.0	114.50	-4.5°	28.71	+40.0°
2.3	3.6	1.0	118.78	-20.0°	24.38	+39.5°
2.3	4.2	-1.0	137.37	+10.0°	31.23	+40.0°
2.3	4.2	0.0	133.68	-3.5°	28.71	+40.0°
2.3	4.2	1.0	137.37	-16.5°	24.38	+39.5°
2.3	4.8	-1.0	156.07	+9.0°	31.23	+40.0°
2.3	4.8	0.0	152.83	-2.5°	28.71	+40.0°
2.3	4.8	1.0	156.07	-14.5°	24.38	+39.5°
5.0	3.6	-1.0	119.56	+15.0°	87.36	+0.0°
5.0	3.6	0.0	115.20	+0.0°	80.00	-35.5°
5.0	3.6	1.0	119.56	-16.0°	87.22	+0.0°
5.0	4.2	-1.0	138.16	+13.0°	87.36	+0.0°
5.0	4.2	0.0	134.40	+0.0°	80.00	-35.5°
5.0	4.2	1.0	138.16	-13.5°	87.22	+0.0°
5.0	4.8	-1.0	156.90	+11.5°	87.36	+0.0°
5.0	4.8	0.0	153.60	+0.0°	80.00	-35.5°
5.0	4.8	1.0	156.90	-12.0°	87.22	+0.0°

4.2 Orthogonal Decoupled Layout (90° Yaw) Sweep

Ankle Spread (E) (cm)	Motor Depth Y-Offset (L) (cm)	Attachment Offset (h) (cm)	Min Pitch Torque (N·m)	Critical Pitch Angle	Min Roll Torque (N·m)	Critical Roll Angle
2.3	3.6	-1.0	81.02	+4.0°	82.22	+6.5°
2.3	3.6	0.0	75.40	-6.5°	75.84	-2.5°
2.3	3.6	1.0	82.09	-16.0°	81.71	-10.5°
2.3	4.2	-1.0	90.41	+6.0°	91.52	+8.0°
2.3	4.2	0.0	85.00	-4.5°	85.42	-2.0°
2.3	4.2	1.0	91.02	-14.5°	90.78	-11.5°
2.3	4.8	-1.0	99.72	+6.5°	100.80	+8.0°
2.3	4.8	0.0	94.56	-3.5°	95.01	-1.5°
2.3	4.8	1.0	100.06	-13.0°	99.90	-11.0°
5.0	3.6	-1.0	102.91	+15.5°	102.91	+16.0°
5.0	3.6	0.0	97.60	+0.0°	97.60	+0.0°
5.0	3.6	1.0	102.89	-16.5°	102.89	-16.0°
5.0	4.2	-1.0	112.24	+13.5°	112.25	+13.5°
5.0	4.2	0.0	107.20	+0.0°	107.20	+0.0°
5.0	4.2	1.0	112.22	-14.0°	112.22	-14.0°
5.0	4.8	-1.0	121.65	+12.0°	121.65	+12.0°
5.0	4.8	0.0	116.80	+0.0°	116.80	+0.0°
5.0	4.8	1.0	121.62	-12.0°	121.62	-12.0°

4.3 Robstride RS04: Peak Sizing Sweep (Limit: 120.0 N·m)

Standard Layout (0° Yaw) Peak Sweep

Spread (E) (cm)	Y-Offset (L) (cm)	Attach (h) (cm)	Min Pitch Torque (N·m)	Min Roll Torque (N·m)
2.3	4.2	-1.0	412.11	93.68
2.3	4.2	0.0	401.05	86.13
2.3	4.2	1.0	412.12	73.13
5.0	4.2	-1.0	414.47	262.07
5.0	4.2	0.0	403.20	240.00
5.0	4.2	1.0	414.47	261.66

Orthogonal Layout (90° Yaw) Peak Sweep

Spread (E) (cm)	Y-Offset (L) (cm)	Attach (h) (cm)	Min Pitch Torque (N·m)	Min Roll Torque (N·m)
2.3	4.2	-1.0	271.22	274.55
2.3	4.2	0.0	255.01	256.27
2.3	4.2	1.0	273.05	272.34
5.0	4.2	-1.0	336.73	336.74
5.0	4.2	0.0	321.60	321.60
5.0	4.2	1.0	336.66	336.65

4.4 RS04 Discrete Joint Angle Torque Matrices (L = 4.2 cm, h = 0.0 cm)

Rated Limit (40.0 N·m) Torque Output

Pitch Angle (Roll=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-25.0°	169.90	187.93	101.80	136.70
-15.0°	140.64	145.81	87.82	113.18
-5.0°	133.81	135.40	85.01	107.71
0.0°	134.11	134.40	85.35	107.20
+5.0°	136.38	135.30	86.59	107.64
+15.0°	147.63	142.57	92.06	111.02
+25.0°	176.46	161.16	105.46	118.79

Peak Limit (120.0 N·m) Torque Output

Roll Angle (Pitch=0°)	Standard E=2.3	Standard E=5.0	Ortho E=2.3	Ortho E=5.0
-30.0°	99.16	240.00	376.48	496.66
-20.0°	105.57	240.00	281.41	356.37
-10.0°	109.21	240.00	260.08	327.99
0.0°	110.16	240.00	256.44	321.60
+10.0°	108.43	240.00	263.53	327.14
+20.0°	103.94	240.00	286.24	346.83
+30.0°	96.56	240.00	365.15	404.91

5. Sizing Conclusions & Optimization Recommendations

Based on the parametric sweeps across rated/peak outputs, ankle spreads, motor depths, and offsets, we conclude:

- **Effect of Spread (E):** Widening the ankle spread from Ankle Spread (E) = 2.3 cm to Ankle Spread (E) = 5.0 cm increases minimum Roll torque by **over 2.6x** in the Standard Parallel layout (from 44.4 N·m to 119.0 N·m peak on RS03) and increases it by **over 1.25x** in the Orthogonal layout.
- **Sensitivity of Depth (L):** Shifting the motor depth Y-offset (L) outwards (e.g. from 3.6 cm to 4.8 cm for RS03/RS04) significantly improves the Pitch torque capability (minimum Pitch torque increases from 113 N·m to 141 N·m peak on Orthogonal RS03). *Recommendation: Keep depth Y-offset (L) as large as allowable by the robot ankle chassis envelope (4.2 cm to 4.8 cm is ideal).*
- **Sensitivity of Attach Offset (h):** Changing attachment offset (h) to +1.0 cm shifts the peak of kinematic transmission efficiency, leading to higher minimum Pitch torques but lower minimum Roll torques. A **nominal h = 0.0 cm** is recommended to maintain balanced, symmetric workspace capacities.
- **Final Verdict:** The **Robstride RS03** is the optimal motor. If a narrow spread E = 2.3 cm must be used, you **MUST** use the **Orthogonal layout** to guarantee roll torque above 100 N·m (gives 112.6 N·m peak). If E = 5.0 cm is feasible, either layout with RS03 is robust, but Orthogonal E = 5.0 cm yields the absolute highest minimum roll capacity (145 N·m peak).